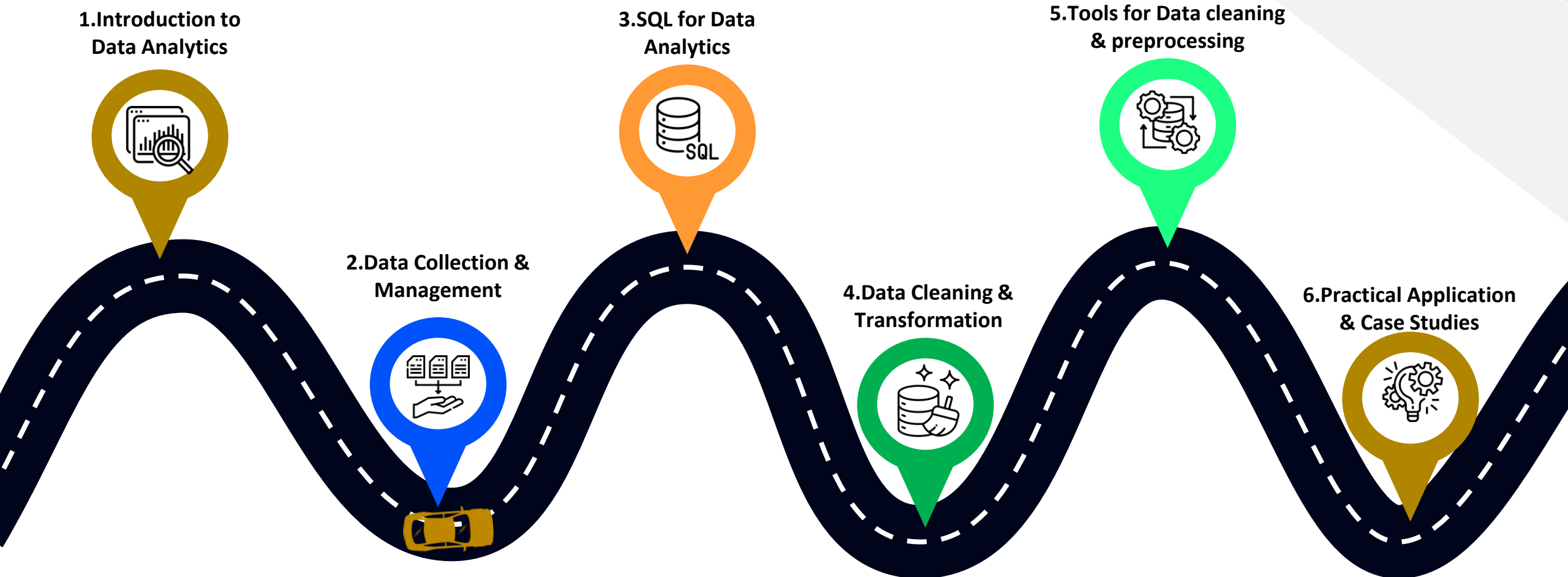




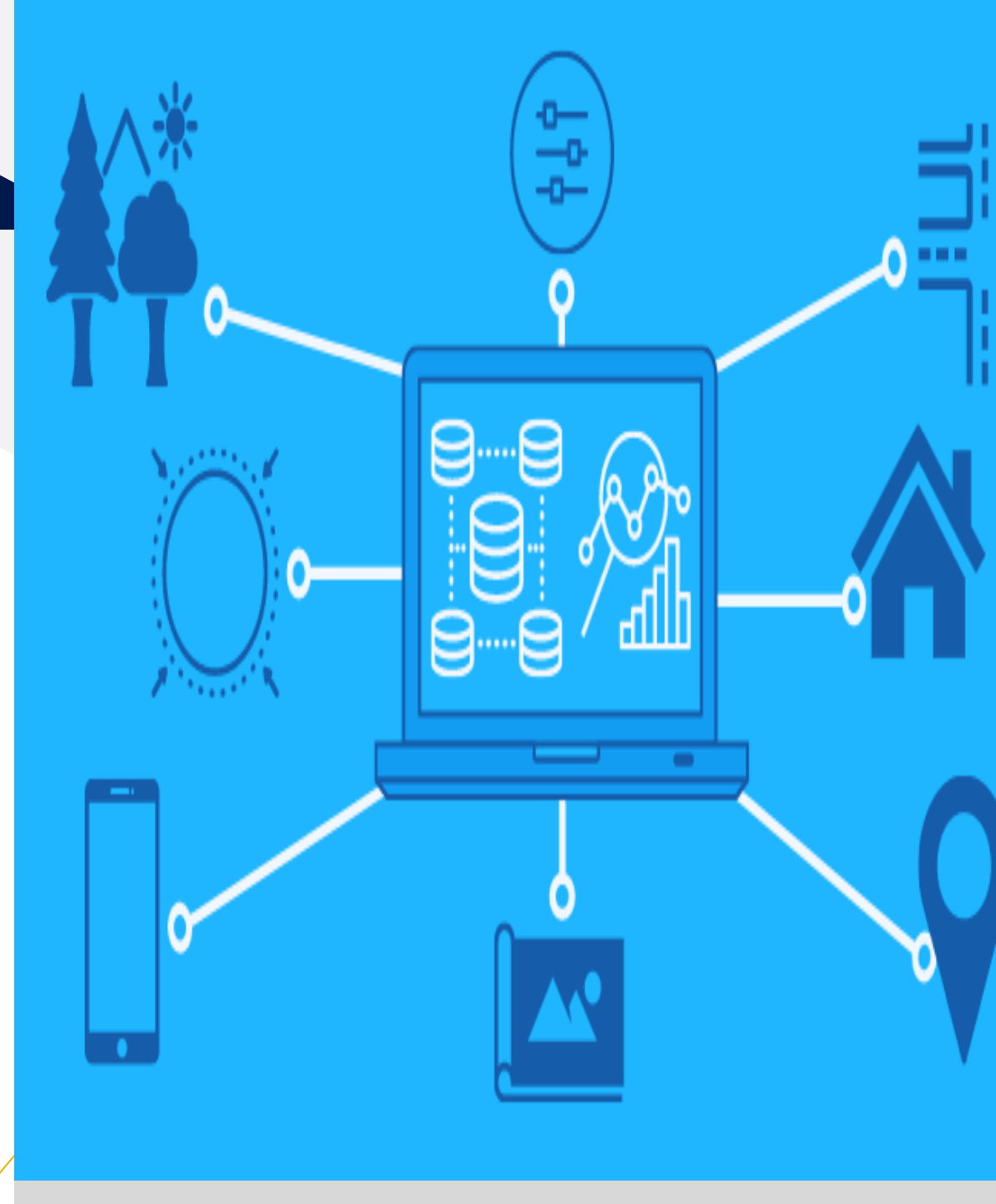
Part Two: Data Collection and Management

Course Road Map



Data Sources

- Internal VS External Sources
- First-Party, Second-Party and Third-Party Data
- Real Time VS Batch Data



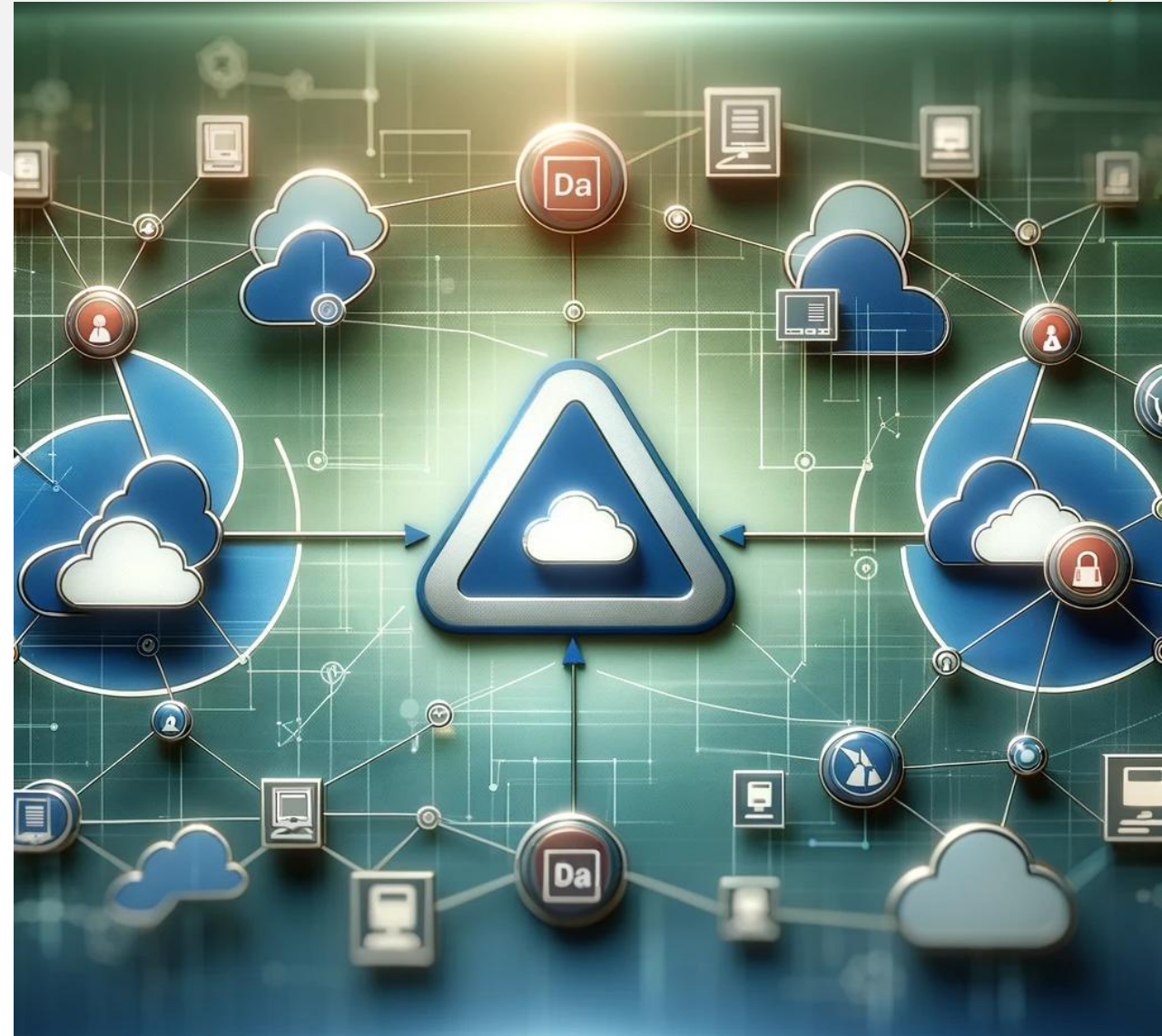
Internal Data Sources

- Company databases, CRM systems, ERP systems.
- Benefits and challenges.



External Data Sources

- Social media, public datasets, market research.
- Benefits and challenges.



First-Party, Second-Party and Third-Party Data

Aspect	First-Party Data	Second-Party Data	Third-Party Data
Definition	Data collected directly from your customers.	Another business's first-party data shared with you.	Data collected by an external entity and sold to businesses.
Examples	Customer names, emails, purchase history.	Data from a partner with a similar audience.	Demographic data from a broad population.
Benefits	Highly accurate and relevant, provides detailed customer insights.	Access to new customer segments, insights from similar audiences.	Large datasets, helps in broad targeting and expanding reach.
Drawbacks	Limited to existing customers, not useful for reaching new audiences.	Potential privacy issues, data quality depends on the partner.	Variable accuracy, potential privacy concerns, less specific to your audience.

Real Time VS Batch Data

Aspect	Batch Data	Real-Time Data
Definition	Data collected and processed in large groups at scheduled intervals.	Data collected and processed continuously as events occur.
Examples	Daily sales reports, monthly financial statements.	Live stock market feeds, real-time social media analytics.
Benefits	Efficient for large volumes of data, lower processing cost.	Immediate insights, suitable for time-sensitive decisions.
Drawbacks	Delay in data availability, not suitable for time-sensitive insights.	Higher processing cost, more complex infrastructure.

Types Of Data Structure

- **Structured Data**
- **Unstructured Data**
- **Semi-Structured Data**



Types Of Data Structure

Aspect	Structured Data	Semi-structured Data	Unstructured Data
Definition	Data organized into rows and columns, easily searchable and analyzable.	Data that has some organizational properties but not a fixed schema.	Data that lacks a predefined format or structure.
Examples	Relational databases, spreadsheets.	JSON files, XML files, email.	Text documents, images, videos, social media posts.
Tools	SQL databases (MySQL, PostgreSQL), Excel.	NoSQL databases (MongoDB, Cassandra), data interchange formats.	Big data tools (Hadoop), text analytics tools, multimedia analysis tools.
Real-World Use Case	Financial transactions, inventory management.	Log files, configuration files, data integration tasks.	Customer feedback analysis, multimedia content management, social media monitoring.

Types Of Data Structure

```

1 <order>
2   <order_id>1024</order_id>
3 <customer>
4   <first_name>Sara</first_name>
5   <last_name>Tarek</last_name>
6 </customer>
7 <items>
8 <item>
9   <product>Cappuccino</product>
10  <quantity>2</quantity>
11  <price>40</price>
12 </item>
13 <item>
14  <product>Croissant</product>
15  <quantity>1</quantity>
16  <price>20</price>
17 </item>
18 </items>
19 <total>100</total>
20 <promo_code_used>true</promo_code_used>
21 </order>

```

XML

```

{
  "order_id": 1024,
  "customer": {
    "first_name": "Sara",
    "last_name": "Tarek"
  },
  "items": [
    {
      "product": "Cappuccino",
      "quantity": 2,
      "price": 40
    },
    {
      "product": "Croissant",
      "quantity": 1,
      "price": 20
    }
  ],
  "total": 100,
  "promo_code_used": true
}

```

JSON

Questions

Data Example	Source	Processing	Structure
Customer purchase records from a company's ERP system	?	?	
Weather data from an API like OpenWeatherMap	?	?	
Comments from your company's Instagram post	?	?	
A survey response form in Excel	?	?	
Emails between the sales team and clients	?	?	
JSON file from a public COVID-19 API	?	?	
Video recordings of customer service calls	?	?	

Databases, APIs, and web scraping

- **Database Key Concepts**
- **Relational Database**
- **Non-Relational Database**
- **APIs & Web Scraping**



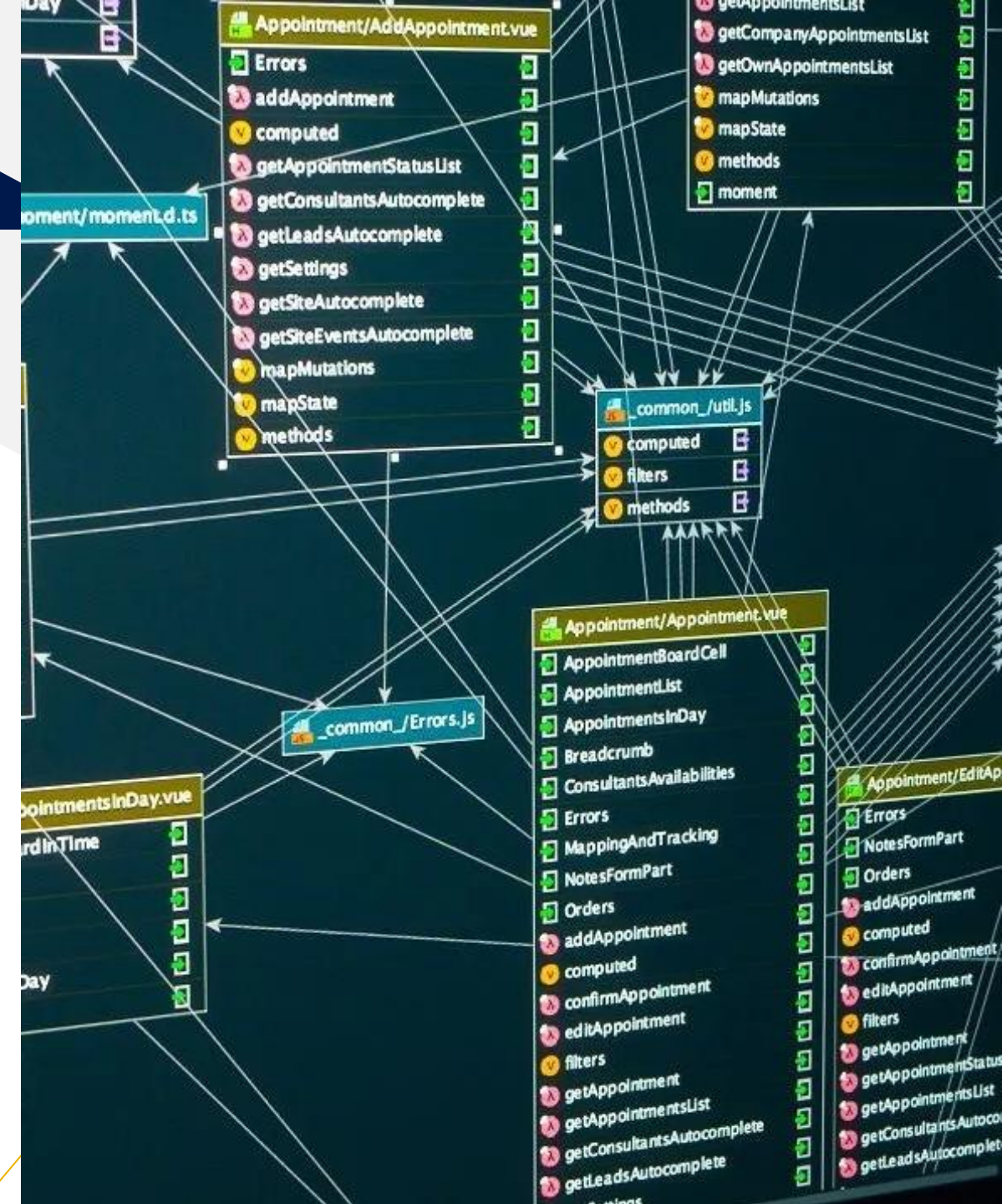
Database Key Concepts

- Database Definition and Main Components
- DBMS and Data Models



Database Definition and Main Components

- A **Database** is an organized collection of data, typically stored and accessed electronically from a computer system. Databases are designed to manage, store, and retrieve large amounts of information efficiently, ensuring data integrity, consistency, and security.



DBMS and Data Models

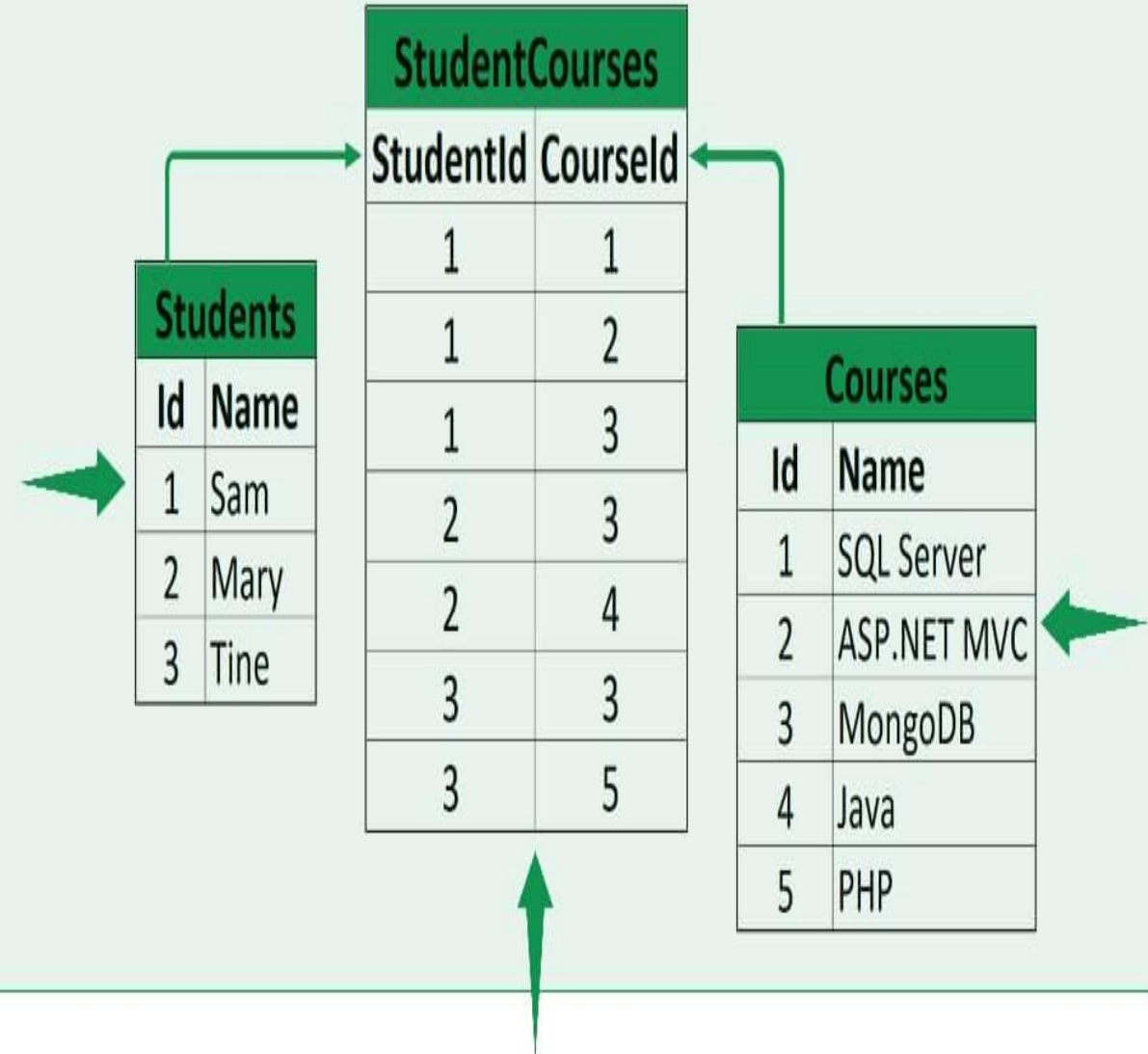
- A **Database Management System (DBMS)** is software that interacts with end-users, applications, and the database itself to capture and analyze data.
- The DBMS provides a systematic way to create, retrieve, update, and manage data. It ensures data integrity, consistency, security, and efficient data management.
- Examples: MySQL, PostgreSQL, Microsoft SQL Server, and MongoDB



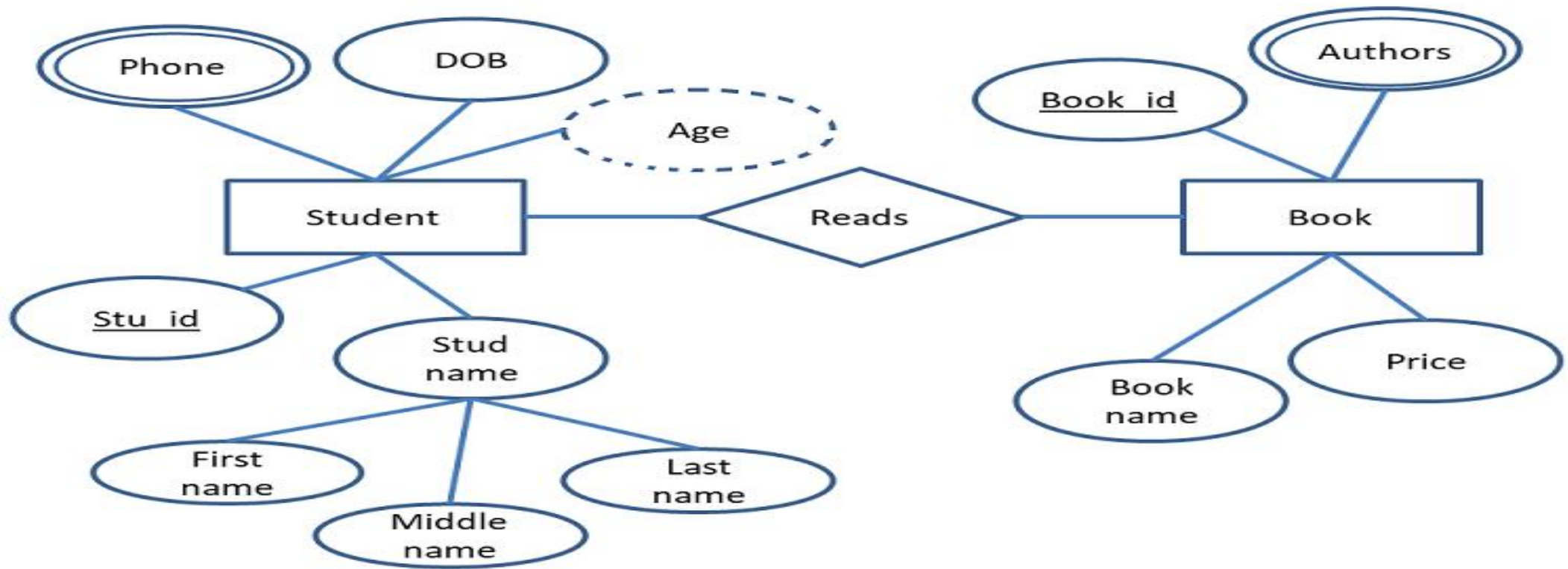
Relational Database

- Entity Relationship Diagram (ERD)
- Relationships Degree
- Cardinality Ratio
- Normalization

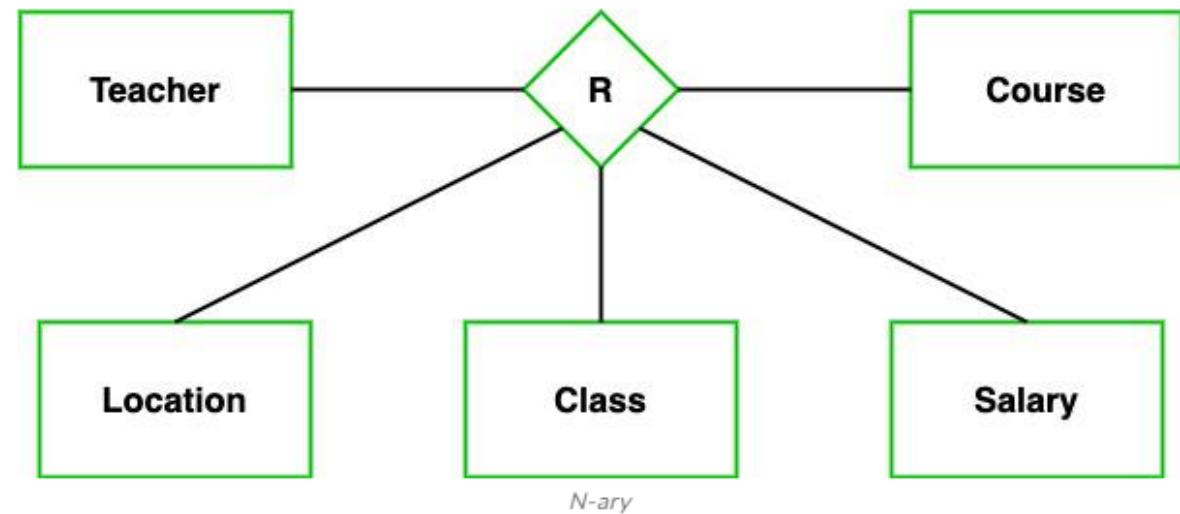
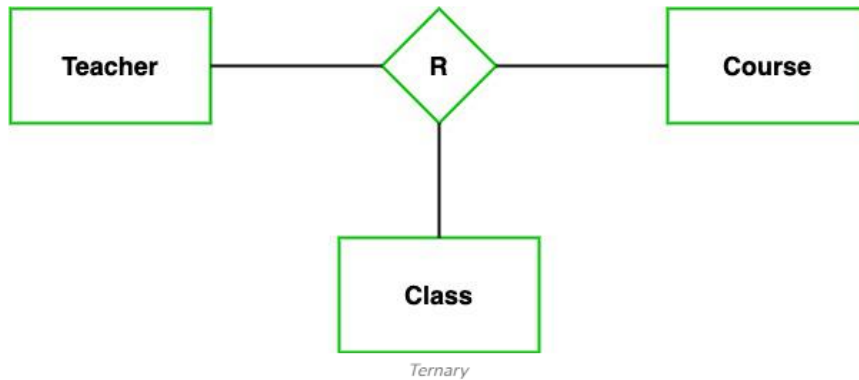
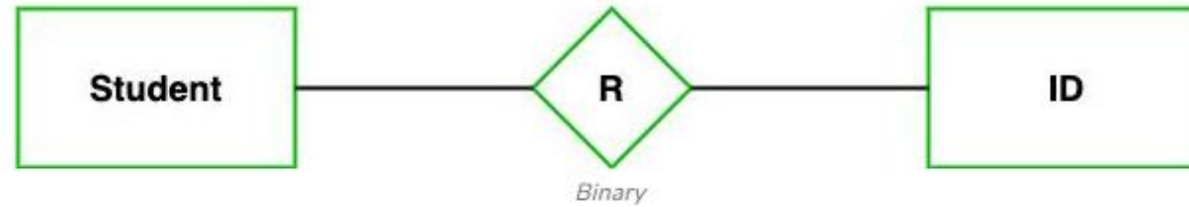
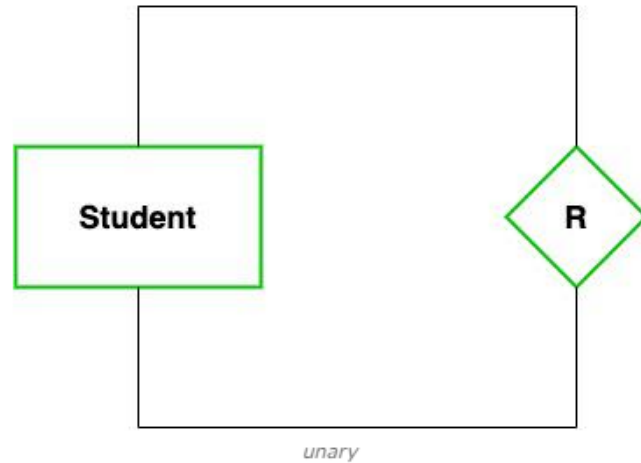
Relational Database



Entity Relationship Diagram (ERD)



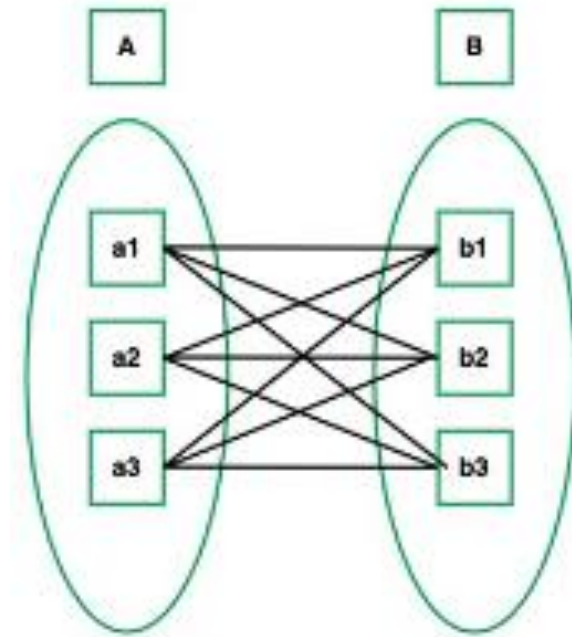
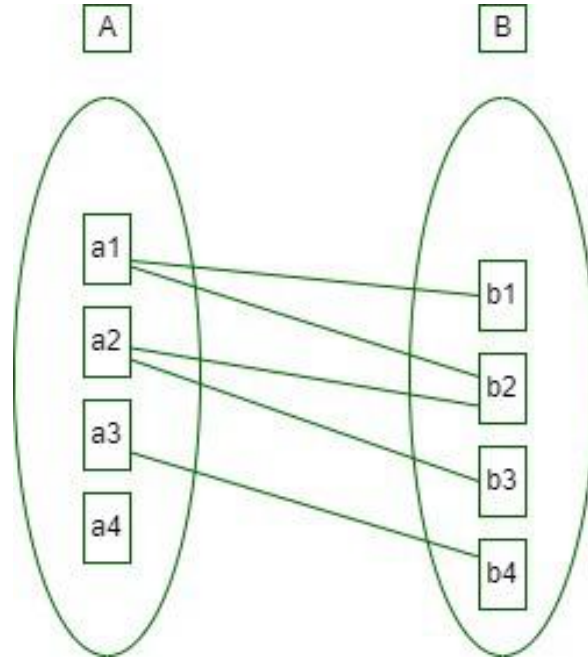
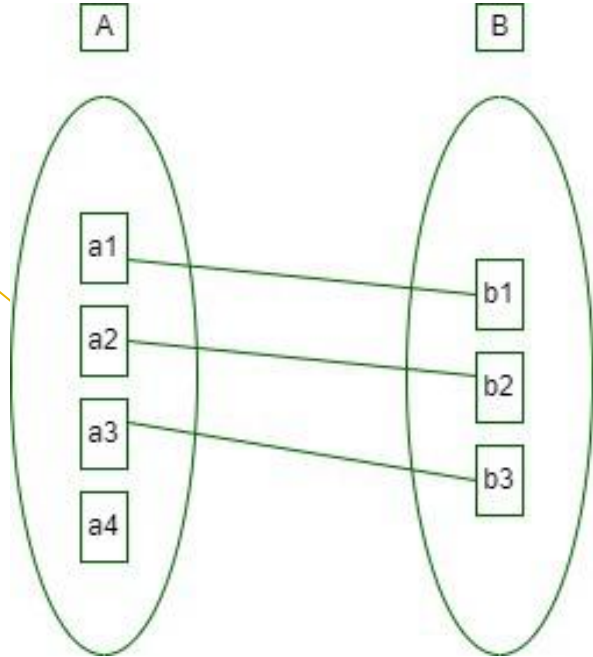
Relationships Degree



Relationships Degree

Aspect	Unary (Degree 1)	Binary (Degree 2)	Ternary (Degree 3)	N-ary (n Degree)
Definition	Relationship within a single entity type	Relationship between two entity types	Relationship involving three entity types	Relationship involving more than three entity types
Examples	Employee manages Employee	Customer places Order	Supplier supplies Product to Warehouse	Company finances Project with Funding from Investors
Use Cases	Self-referencing relationships	Common in relational databases	Complex relationships involving three entities	Advanced database schemas involving multiple entities

Cardinality Ratio



Cardinality Ratio

- One to One Relationship
- One to Many Relationship
- Many to many Relationship

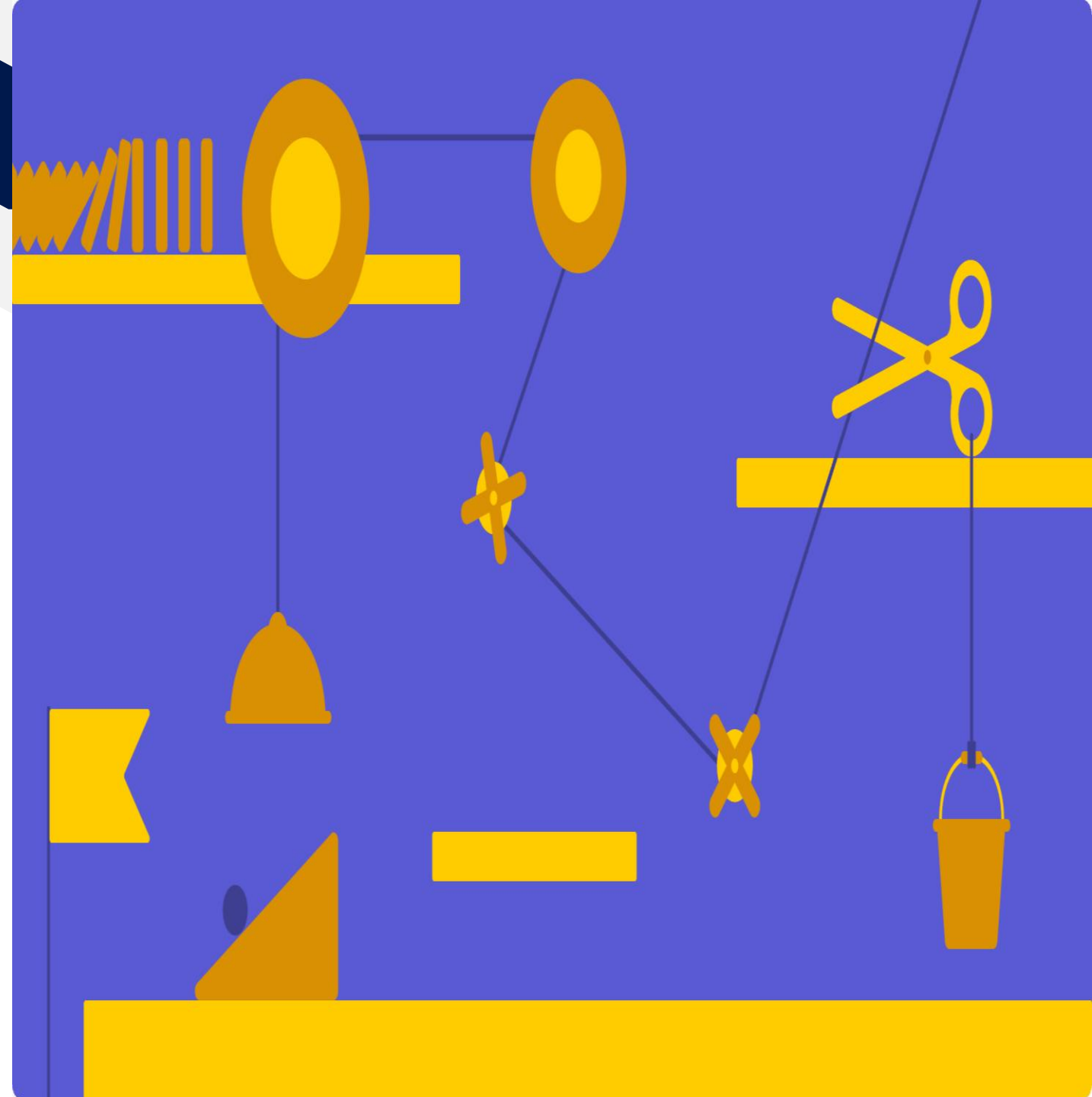


Case Study



Normalization

- What is Normalization
- Functional Dependency
- Rules of 1st Normal Form
- Rules of 2nd Normal Form
- Rules of 3rd Normal Form



What is Normalization

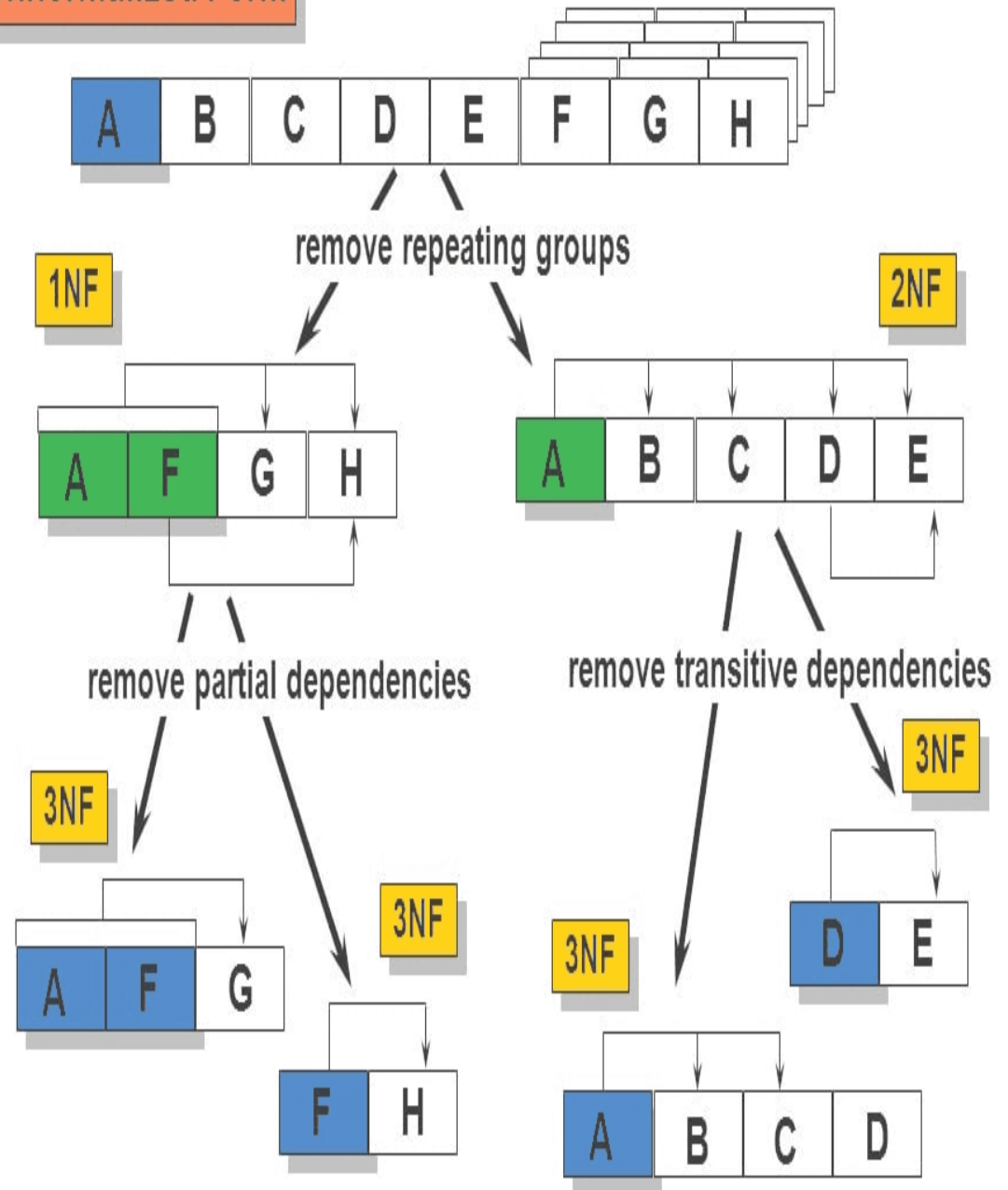
- Normalization is a process in database design that organizes tables and attributes to reduce redundancy and dependency.
- It typically involves multiple phases, commonly referred to as Normal Forms (NF).



Functional Dependency

- Functional dependency describes the relationship between attributes in a relational database.
- It indicates that the value of one attribute (or a group of attributes) determines the value of another attribute.

Unnormalized Form



Normalization Roles

Normal Form	Key Characteristics	Primary Objective
1NF (First Normal Form)	- Eliminates repeating groups. - Ensures atomicity of columns (each column has a single value). - Identifies a primary key for each row.	To eliminate redundancy and ensure each column contains only atomic values.
2NF (Second Normal Form)	- Removes partial dependencies. - Ensures non-key attributes are fully functionally dependent on the primary key. - Non-key attributes dependent on part of the primary key are moved to separate tables.	To remove redundancy and ensure all non-key attributes are dependent only on the whole primary key.
3NF (Third Normal Form)	- Eliminates transitive dependencies. - Ensures non-key attributes are not dependent on other non-key attributes. - Non-key attributes dependent on other non-key attributes are moved to separate tables.	To further reduce redundancy and ensure data integrity by removing indirect relationships between non-key attributes.

Normalization Examples

Sales Table (ONF)

SaleID	Customer Name	Phone	Product	Category	Price	Date
1	Ahmed	01012345678	Laptop, Mouse	Computer	1200	2025-06-01
2	Sara	01087654321	Phone	Mobile	900	2025-06-02
3	Ahmed	01012345678	Laptop	Computer	1200	2025-06-03

Normalization Examples

Sales Table (ONF)

SaleID	CustomerName	Phone	Product	Category	Price	Date
1	Ahmed	01012345678	Laptop	Computer	1200	2025-06-01
1	Ahmed	01012345678	Mouse	Computer	50	2025-06-01
2	Sara	01087654321	Phone	Mobile	900	2025-06-02
3	Ahmed	01012345678	Laptop	Computer	1200	2025-06-03

Normalization Examples

Sales Details (1NF)

SaleID	Product	Category	Price
1	Laptop	Computer	1200
1	Mouse	Computer	50
2	Phone	Mobile	900
3	Laptop	Computer	1200

Sales Table (1NF)

SaleID	CustomerName	Phone	Date
1	Ahmed	01012345678	2025-06-01
2	Sara	01087654321	2025-06-02
3	Ahmed	01012345678	2025-06-03

Normalization Examples

Sales Details (1NF)

SaleID	Product	Category	Price
1	Laptop	Computer	1200
1	Mouse	Computer	50
2	Phone	Mobile	900
3	Laptop	Computer	1200

Sales Table (1NF)

SaleID	CustomerName	Phone	Date
1	Ahmed	01012345678	2025-06-01
2	Sara	01087654321	2025-06-02
3	Ahmed	01012345678	2025-06-03

Normalization Examples

Sales Details (2NF)

SaleID	Product
1	Laptop
1	Mouse
2	Phone
3	Laptop

Products Table (2NF)

Product	Category	Price
Laptop	Computer	1200
Mouse	Computer	50
Phone	Mobile	900

Sales Table (2NF)

SaleID	CustomerName	Phone	Date
1	Ahmed	01012345678	2025-06-01
2	Sara	01087654321	2025-06-02
3	Ahmed	01012345678	2025-06-03

Normalization Examples

Sales Details (3NF)

SaleID	Product
1	Laptop
1	Mouse
2	Phone
3	Laptop

Products Table (3NF)

Product	Category	Price
Laptop	Computer	1200
Mouse	Computer	50
Phone	Mobile	900

Sales Table (3NF)

SaleID	CustomerID	Date
1	C001	2025-06-01
2	C002	2025-06-02
3	C001	2025-06-03

Customers Table (3NF)

CustomerID	CustomerName	Phone
C001	Ahmed	01012345678
C002	Sara	01087654321

Non-Relational Database

- Examples and use cases



Examples and Use Cases

Non-Relational Database	MongoDB	Cassandra	Redis	Neo4j
Type	Document-oriented	Wide-column store	Key-value store	Graph database
Examples	JSON-like documents, collections	Tables with flexible schemas	Key-value pairs	Nodes, relationships, properties
Use Cases	CMS, real-time analytics, mobile applications	High-availability applications, real-time big data, IoT applications	Caching, session management, real-time analytics	Social networks, fraud detection, recommendation engines

APIs & Web Scraping

- APIs Definition
- APIs Examples: Twitter API, Google Maps API.
- How to connect to an API and retrieve data.
- Python Application (Web Scraping)
- Ethical and legal considerations.



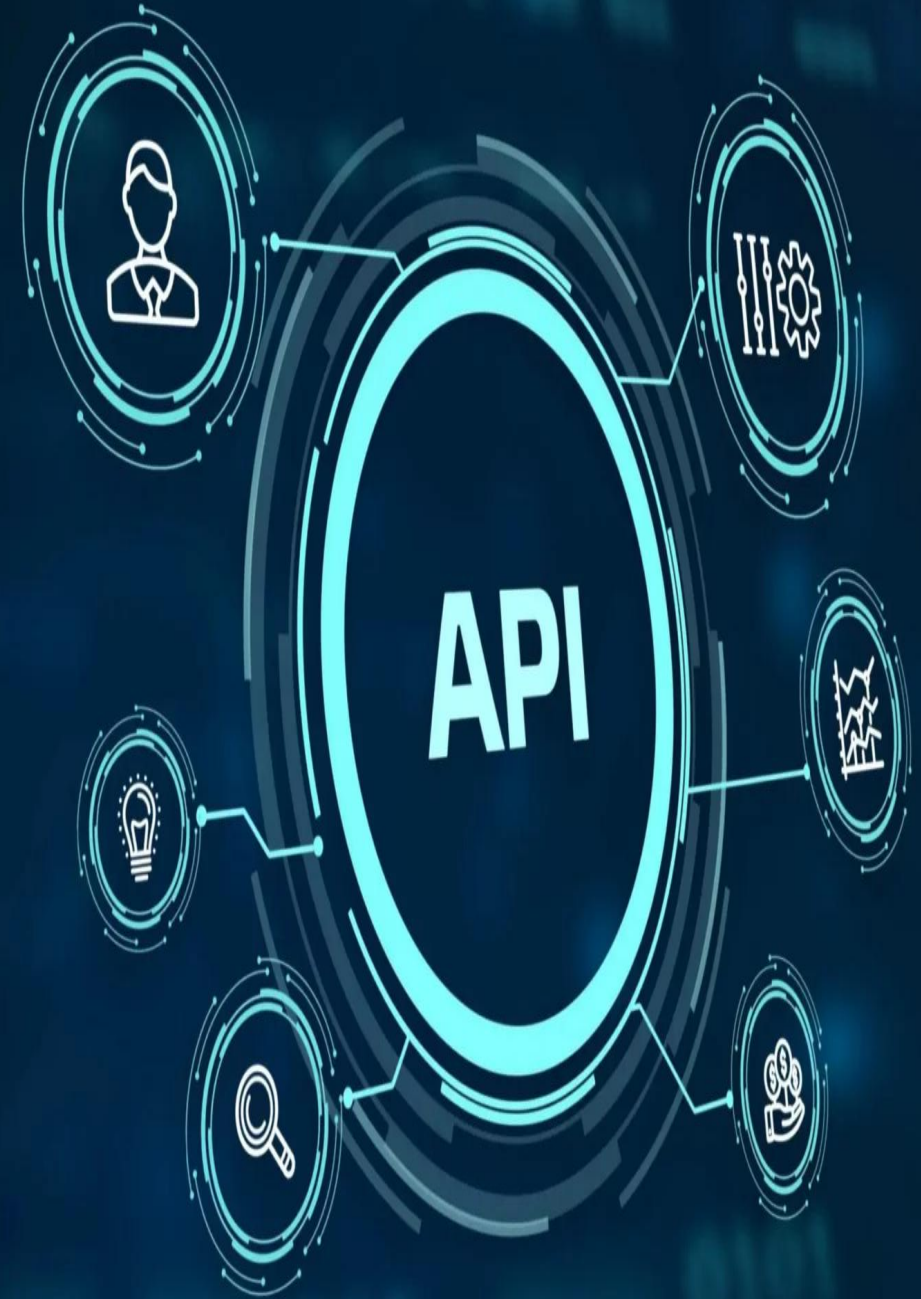
APIs

- An API, or Application Programming Interface, is a set of rules and protocols for building and interacting with software applications.
- It defines the methods and data formats that applications can use to communicate with each other.

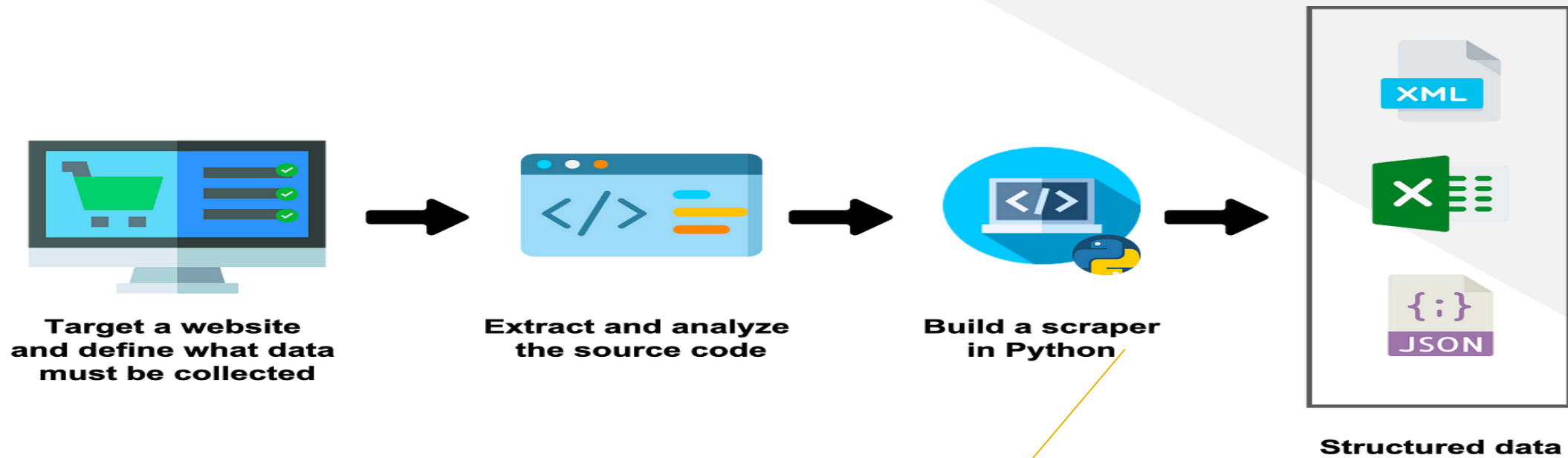


Types of APIs

- Web APIs:
 - REST (Representational State Transfer)
 - SOAP (Simple Object Access Protocol)
- Library-Based APIs:
 - Sets of routines, protocols, and tools for building software and applications.
 - For example, the standard libraries in languages like Java, C++, or Python.



(Web Scrapping with Python)



Questions





Thank You.



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